WORKSHEET

Chapter 9: Allocation and the Hungarian algorithm

1 Draw a bipartite graph for each of the following scenarios.

a Three housemates are sharing cleaning tasks as follows.

- Peta will clean the kitchen or do the vacuuming.
- Michael will clean the kitchen or the bathroom.
- Lazlo will do any of the three tasks.

b Four teachers teach different subjects as follows.

- Alison teaches English and History.
- Mario teaches History and Business Studies.
- Bob teaches PDHPE, Maths and Science.
- Janice teaches Maths and Science.

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c Four students represent their school in the following sports.

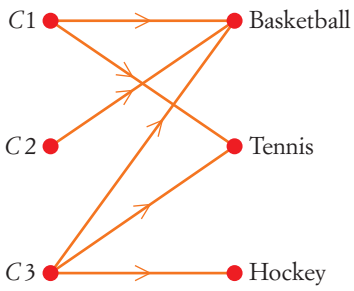
Student	Sport
Brittany	Basketball, netball
Charlie	Hockey, basketball, soccer
Nazim	Hockey
Pearl	Soccer, netball, basketball

d There are four players in a tennis round-robin competition. Each player must play every other player once.

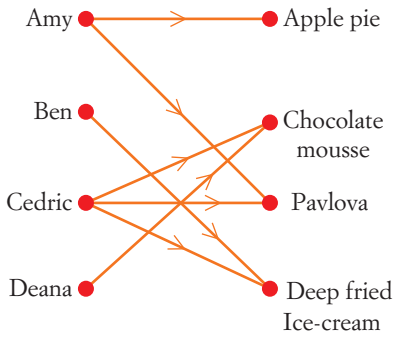
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2 Determine allocations using each of the following bipartite graphs.

a Three coaches coach basketball, tennis and hockey.

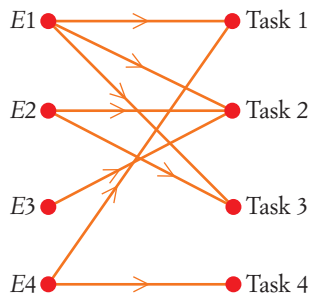


b Four different desserts are shared between four friends.

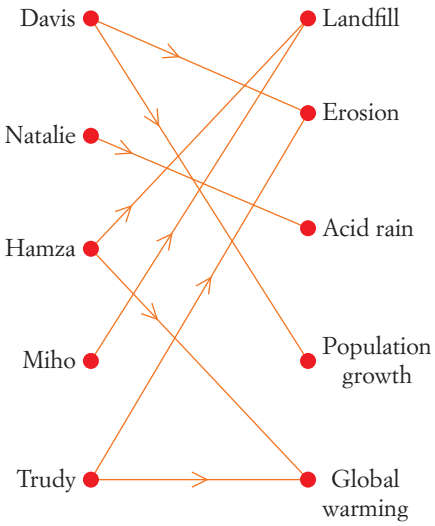


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c Four employees are allocated four different tasks.



d Five Geography students present to the class on five different topics.



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3 The table below shows quotes (in \$1000s) from three companies *A*, *B* and *C* for three different jobs.

	1	2	3
A	15	6	8
B	14	19	12
C	12	15	11

a Find the table that results from a row and column reduction.

b Draw a bipartite graph showing all the connections indicated in part **a**.

c State the optimum allocation.

d Find the total cost for all three jobs based on your answer to part **c**.

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4 Three workers on a construction site are to be assigned three different tasks. The times (in minutes) each worker takes for each task are recorded in the table below.

	*T*1	*T*2	*T*3
Johan	31	38	29
Trent	45	41	40
Abi	26	22	23

Use row and column reduction to determine the optimum allocation to minimise the time spent on the three tasks.

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5 The times taken for three nurses (A, B, C) to carry out three different procedures (1, 2, 3) are recorded in the following table.

	1	2	3
A	7	3	8
B	5	6	9
C	10	5	12

a Find the table that results from a row and column reduction.

b Use the Hungarian algorithm to produce a table from which an optimum allocation can be made.

c State the optimum allocation.

d Calculate the total working time for the three procedures.

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6 A taxi company has four taxis available, *A*, *B*, *C* and *D,* and there are four customers requiring a taxi each. The distance (km) that each taxi must travel to reach the customers is shown in the following table.

	1	2	3	4
A	12	13	6	22
B	11	15	13	23
C	23	22	12	15
D	19	17	6	21

a Use the Hungarian algorithm to find the optimum allocation of taxis to customers that minimises the distance travelled.

b If this allocation is implemented, find the total distance travelled.

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7 The annual cost of shipping goods (in \$1000s) from four warehouses W, X, Y and Z to four different towns is shown in the following table.

	1	2	3	4
W	18	51	33	26
X	27	55	35	19
Y	19	46	25	22
Z	23	66	41	17

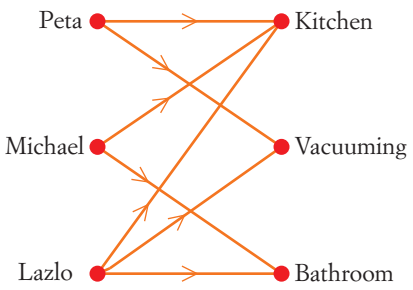
a Use the Hungarian algorithm to find the optimum allocation that minimises shipping costs.

b Find the total annual cost of shipping the goods using this allocation.

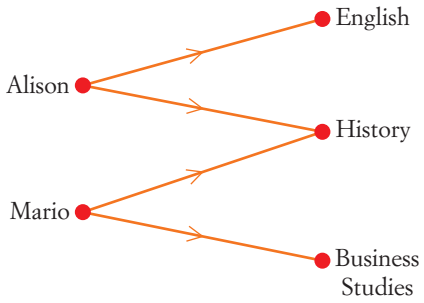
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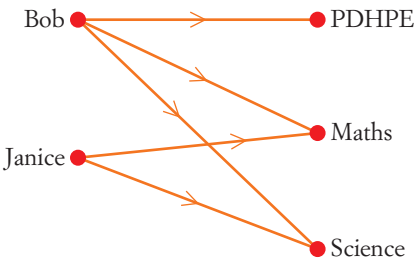
Answers

1 a

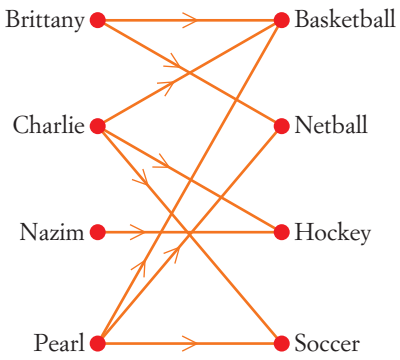


b

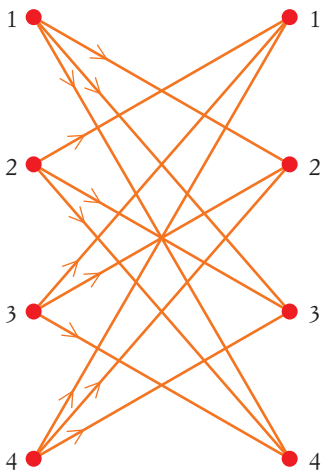




c



d



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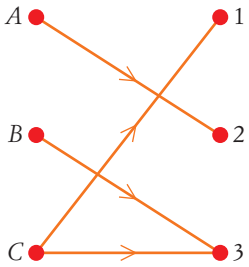
2 **a** $C1$: Tennis; $C2$: Basketball; $C3$: Hockey.

b Amy: Apple pie; Ben: Deep fried ice-cream; Cedric: Pavlova; Deana: Chocolate mousse.

c $E1$:Task 1; $E2$: Task 3; $E3$: Task 2; $E4$: Task 4.

d Davis: Population growth; Natalie: Acid rain; Hamza: Global warming; Miho: Landfill; Trudy: Erosion.

3 **a**
$$\begin{bmatrix} 8 & 0 & 2 \\ 1 & 7 & 0 \\ 0 & 4 & 0 \end{bmatrix}$$

b


c $A2$, $B3$, $C1$

d \$30 000

4 Johan: $T1$, Trent: $T3$, Abi: $T2$.

5 **a**
$$\begin{bmatrix} 4 & 0 & 1 \\ 0 & 1 & 0 \\ 5 & 0 & 3 \end{bmatrix}$$

b
$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

c $A3$, $B1$, $C2$

d 18 minutes

6 **a** $A2$, $B1$, $C4$, $D3$

b 45 km

7 **a** $W1$, $X2$, $Y3$, $Z4$

b \$115 000

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